POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Physics

Semester: Fall

Year : 2021
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Why is SHM important in physics? You are given a compound pendulum. How can you show that it oscillates simple harmonically? Hence find its time period. Also show that point of suspension and point of oscillation are interchangeable. 3+3+3

   b) Here are the three equations of a wave: 3+3

   $y(x,t) = 2\sin(4x-2t)$, $y(x,t) = 2\sin(3x-4t)$, $y(x,t) = 2\sin(3x-3t)$.

   Rank the waves according to their (a) wave speed and (b) maximum transverse speed with greatest first.

2. a) Define interference of light wave. Derive the theory of Newton's Ring experiment to find the value of wavelength of mono-chromatic light wave. 9

   b) What is reverberation and reverberation time? Derive Sabine's formula. 6

3. a) Define the term capacitance. Hence derive the expression considering of charging and discharging of capacitor through resister. 9

   b) Deduce the missing order for a double slit fraunhoffer's diffraction pattern if the slit widths are 0.16mm and they are 0.8mm apart. 6

4. a) Compare Ampere's law with Biot-Savart law. Use Biot-Savart law to calculate the magnitude of magnetic field strength due to current carrying solenoid. 2+7

   b) Find electric field intensity at the center of a square of side 5cm consisting of 2μFcharges in each vertex. 6

5. a) What are Maxwell's equation? Convert the integral form of Maxwell's equation in its differential form, with physical significance. 9

   b) What are the (i) mean time between collisions? (ii) The mean free path for free electrons in copper? (Given $n=8.4\times10^{28}m^{-3}$, $\rho=1.7\times10^{-8}\Omega m$, $v_{avg}=1.6\times10^{6}$ m/s.) 6

6. a) What is the significance of wave function ψ? Derive the time independent Schrodinger wave equation for a free particle. 9

   b) Using Maxwell's equation, prove that $C=\frac{E_0}{B_0}$ Where symbols carry their usual meaning. 6

7. Write short notes on: (**Any two**) 2×5

   a) LASER and its application

   b) PN junction and its biasing

   c) Magnetic energy density